

JIHO CHOI

최지호

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EDUCATION

M.S. in Smart City Engineering • *Inha University, Incheon, Korea*

Expected Graduation: February 2027 • GPA: 4.0/4.0

B.S. in Civil Engineering • *Inha University, Incheon, Korea*

Graduated: February 2025 • GPA: 3.08/4.0

RESEARCH EXPERIENCE

Prediction of Spatiotemporal Nonlinear Temperature Distribution of Airport Pavement using AI Learning of Long-term Field Measurement Data, NRF, Sep 2025 – Aug 2027.

- Developing AI-based predictive models using long-term field measurement data for nonlinear temperature distribution analysis in airport pavements

Development of Technologies for Reducing Road Surface Hazards and Evaluating Road Safety Levels, KAIA, Apr 2023 – Dec 2027.

Analysis of Snow Removal and Freezing Vulnerability Factors in Seoul and Derivation of Selection Condition Systems, The Seoul Institute / KSRE, Feb 2026 – Aug 2026.

- Analyzing road surface hazards and developing technologies for evaluating road safety levels using GIS and statistical methods
- Assessing snow removal and freezing vulnerability factors in Seoul using DEM (Digital Elevation Models) and spatial analysis

Structural Analysis and Statistical Analysis Considering the Influence of Traffic and Environmental Loads on Cement-based Bridge Deck Pavement, Korea Expressway Corporation Research Institute, Jun 2025 – Dec 2026.

- Conducting nonlinear FEA / Machine Learning to simulate pavement structure behavior under traffic and environmental loads

PUBLICATIONS & PRESENTATIONS

- Choi, J.H., Park, G., Jeong, J.H. (2026). *Estimation of Highway Geometric Features Using Public Digital Elevation Models (DEM) and GIS Road Network Data*. Proceedings of KSRE 2026 Spring Conference. BEST PAPER AWARD
- Choi, J.H., Song, K.Y., Son, S.C., Noh, G.H., Kim, S.Y., Jeong, J.H. (2025). *Estimation of Highway Geometric Features from Digital Elevation Models and Node-Link Data*. Proceedings of 2025 International Conference for Road Engineers, pp. 102–103.
- Choi, J.H., Kim, S.H., Son, S.C., Park, G., Jeong, J.H. (2025). *Estimation of Road Geometric Features Based on Horizontal Alignment*. Proceedings of KSRE 2025 Spring Conference, pp. 123–124.
- Lee, J.H., Choi, J.H., Baek, S.B., Son, S.C., Song, K.Y., Park, G., Jeong, J.H. (2024). *Study on the Calculation of Stress Relief Joints for Concrete Pavements Considering Regional Characteristics in Korea*. Proceedings of KSRE 24th Annual Conference, pp. 39–40.
- Woo, B.C., Kim, Y.T., Kim, J.S., Choi, J.H., Jeong, J.H. (2022). *Correlation Analysis between Pavement Distress Factors and Influence Factors on National Highways in Gangwon-do*. Proceedings of KSRE 2022 Spring Conference, p. 25.

ACADEMIC HONORS & ROLES

- Best Case Award, Graduate School Excellence Case Contest (Inha University), January 2026
- Graduate Research & Teaching Assistant, Inha University (2025–2027)
- Academic Merit Scholarship, Department of Civil Engineering, Inha University (2021–2023)

TECHNICAL SKILLS

- Finite Element Analysis (FEA): ABAQUS (Concrete Damage Plasticity, CDP model)
- GIS & Spatial Analysis: QGIS, Digital Elevation Models (DEM)
- Machine Learning & Data Analysis: Python, Random Forest algorithms
- Design & Visualization: AutoCAD, Revit 2025